

Lab 4: VibeCoding — Complete Chat History

Student: Rohan Chukkapalli (PES1UG24CS135) | **Repo:** VortexVector/09_tic_tac_toe

User

https://github.com/SETAPESU26/09_tic_tac_toe

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:55:01+05:30

Completed At: 2026-09-30T11:55:17+05:30

Title: Live Content

OG Description: Fetched live

The full content of the article at https://raw.githubusercontent.com/SETAPESU26/09_tic_tac_toe/main/README.md has been saved to:
C:\Users\rchuk\gemini\antigravity\brain\797e1d1a-07d3-4d1b-a401-45de60e4324a\system_generated\steps\2\content.md

You can use the `view_file` tool to read specific sections if needed.

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:55:24+05:30

Completed At: 2026-09-30T11:55:25+05:30

File Path: `file:///C:/Users/rchuk/gemini/antigravity/brain/797e1d1a-07d3-4d1b-a401-45de60e4324a/system\_generated/steps/2/content.md`

Total Lines: 84

Total Bytes: 2235

Showing lines 1 to 84

The following code has been modified to include a line number before every line, in the format: `<line_number>: <original_line>`. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

1: Title: Live Content

2:

3: Description: Fetched live

4:

5: Source: https://raw.githubusercontent.com/SETAPESU26/09_tic_tac_toe/main/README.md

6:

7: ---

8:

9: # Tic-Tac-Toe Lab

10:

11: This project is a single-topic two-player Tic-Tac-Toe game using

12: `**Pygame**`. It introduces students to win-condition logic, turn/move

13: validation, and persistent state across rounds, using a small,

14: readable object-oriented codebase.

15:

16: ---

17:

18: ## What's Provided

19:

20: A working Tic-Tac-Toe game with:

21:

22: - A 3x3 board you click on to place X

23: - You always play X; the computer automatically plays O right after

24: you, using a simple random-move opponent - this lab is designed for

25: one person to play solo against the computer, not for two people

26: sharing a keyboard

27: - Basic win and draw detection, and a "Press R for a new round"

28: restart

29:
30: It has **one deliberate bug** (with several related symptoms) and
31: **three features** left for you to build. You are expected to
32: **analyze**, **interact with an AI assistant**, and **complete/fix**
33: the game to make it fully functional and more interesting.
34:
35: **Use an LLM (e.g. ChatGPT or Claude) as your debugging and pair-programming partner for this lab.**
36:
37: ---
38:
39: **Getting Started**
40:
41: **Setup**
42:
43: 1. Make sure you have Python 3.10+ installed.
44: 2. Install dependencies:
45:
46: ```bash
47: pip install -r requirements.txt
48: ```
49:
50: 3. Run the game:
51:
52: ```bash
53: python main.py
54: ```
55:
56: **Controls:** You play X - click a cell to place it. The computer
57: automatically plays O right after you. Press R for a new round.
58:
59: ---
60:
61: **Tasks to Complete**
62:
63: Each task must be completed using an iterative process involving LLM
64: suggestions and your critical code review.
65:
66: **Task 1: Fix win and draw detection**
67:
68: > **What you'll see, problem 1:** win three-in-a-row diagonally (top
69: > corner to bottom corner, either direction) and the game doesn't
70: > notice at all - no winner is announced, even though the board
71: > clearly shows one.
72:
73: > **What you'll see, problem 2:** if the winning move also happens to
74: > be the move that fills the very last empty cell, the game announces
75: > "Draw!" instead of announcing the actual winner.
76: >
77:
78: > **What you'll see, problem 3:** once a round has ended, you can keep
79: > clicking empty cells and new symbols keep appearing on the board.
80: >
81:
82: > **Fix all three**
83:
84:

The above content shows the entire, complete file contents of the requested file.

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:55:30+05:30

Completed At: 2026-09-30T11:55:48+05:30

The command exited with code 0.

Output:

Directory: C:\Users\rchuk\Downloads\files (2)

Mode LastWriteTime Length Name

```
----
d----- 22-04-2026 11:48 screenshots
-a---- 10-04-2026 12:01 10225 controller.py
-a---- 10-04-2026 12:01 7890 README.md
-a---- 30-09-2026 11:51 4497 readme_fetched.md
-a---- 10-04-2026 12:01 4695 test_scenarios.sh
-a---- 10-04-2026 12:01 9027 topology.py
```

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:55:51+05:30

Completed At: 2026-09-30T11:55:51+05:30

File Path: `file:///c:/Users/rchuk/Downloads/files%20%282%29/readme\_fetched.md`

Total Lines: 145

Total Bytes: 4497

Showing lines 1 to 145

The following code has been modified to include a line number before every line, in the format: <line_number>: <original_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: # Tic-Tac-Toe Lab
2:
3: This project is a single-topic two-player Tic-Tac-Toe game using
4: **Pygame**. It introduces students to win-condition logic, turn/move
5: validation, and persistent state across rounds, using a small,
6: readable object-oriented codebase.
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8: ---
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12: A working Tic-Tac-Toe game with:
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14: - A 3x3 board you click on to place X
15: - You always play X; the computer automatically plays O right after
16: you, using a simple random-move opponent - this lab is designed for
17: one person to play solo against the computer, not for two people
18: sharing a keyboard
19: - Basic win and draw detection, and a "Press R for a new round"
20: restart
21:
22: It has **one deliberate bug** (with several related symptoms) and
23: **three features** left for you to build. You are expected to
24: **analyze**, **interact with an AI assistant**, and **complete/fix**
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67: > "Draw!" instead of announcing the actual winner.
68: >
69:
70: > **What you'll see, problem 3:** once a round has ended, you can keep
71: > clicking empty cells and new symbols keep appearing on the board.
72: >
73:
74: > **Fix all three:** diagonals should count as wins, a genuine win
75: > should always be reported as a win even if the board is also full,
76: > and no further moves should be accepted once a round has ended.
77:
78: ### Task 2: Implement a persistent scoreboard
79:
80: > Track how many rounds X has won, how many O has won, and how many
81: > have ended in a draw. This should stay visible and unchanged when
82: > starting a new round, and should only be cleared when the player
83: > explicitly resets the whole match (not just the round).
84:
85: ### Task 3: Implement turn and move validation
86:
```

87: > ****What you'll see:**** click on a cell that already has a symbol in
88: > it, and your click overwrites it with your own symbol instead of
89: > being rejected.
90: >
91:
92: > ****Fix it:**** a click on an already-occupied cell should be rejected
93: > entirely - the board and whose turn it is should both stay exactly
94: > as they were.
95:
96: **### Task 4: Add first-player choice and separate restart controls**
97:
98: > Let the player choose whether X or O goes first (since the computer
99: > always plays O, choosing "O starts" means the computer takes the
100: > first move of the round automatically). Provide two separate
101: > controls: one to restart just the current round (keeping the
102: > scoreboard), and one to reset the whole match (clearing the
103: > scoreboard too).
104:
105: ---
106:
107: **## Expected Behavior**
108:
109: - All 8 winning lines - 3 rows, 3 columns, and both diagonals - are
110: detected correctly.
111: - A move that both wins the game and fills the last empty cell is
112: always scored as a win, never as a draw.
113: - Once a round has ended, clicking anywhere on the board does nothing.
114: - Clicking an already-occupied cell is rejected outright - the board
115: and current turn stay unchanged.
116: - The scoreboard survives a round restart but resets to zero on a full
117: match reset.
118: - Choosing which symbol starts takes effect on the next round.
119:
120: ---
121:
122: **## Folder Structure**
123:
124: ```
125: tic-tac-toe/
126: ■■■ main.py
127: ■■■ requirements.txt
128: ■■■ game/
129: ■ ■■■ game_engine.py
130: ■ ■■■ rules.py
131: ■ ■■■ ai.py
132: ■ ■■■ renderer.py
133: ■■■ README.md
134: ```
135:
136: ---
137:
138: **## Submission Checklist**
139:
140: Submission is only the following three things:
141:
142: - [] A 10-second video of gameplay ****before**** your changes, showing the bug/broken behavior
143: - [] A 10-second video of gameplay ****after**** your changes, showing the bug fixed and the new features working
144: - [] The Chat/LLM used page link, with the complete chat history

145:

The above content shows the entire, complete file contents of the requested file.

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:55:58+05:30

Completed At: 2026-09-30T11:56:12+05:30

The command exited with code 0.

Output:

<truncated 491 lines>

```
-a---- 26-06-2025 15:38 221184 UE24EE141B_ec12c54a-eba5-11ef-b61c-02e014bc5e7b_20250222101402.pptx
-a---- 06-04-2025 09:47 3031040 UE24EV121A_8d24d4fb-6f60-11ef-b7ef-02bdb07e3b51_20240918140411.pptx
-a---- 06-04-2025 09:40 17289216 UE24EV121A_8d24d57d-6f60-11ef-b7ef-02bdb07e3b51_20240918135246(1).ppt
x
-a---- 15-02-2025 10:26 17289216 UE24EV121A_8d24d57d-6f60-11ef-b7ef-02bdb07e3b51_20240918135246.pptx
-a---- 06-04-2025 09:46 44150784 UE24EV121A_8d24d5b9-6f60-11ef-b7ef-02bdb07e3b51_20240918140122.pptx
-a---- 12-06-2025 17:50 17137664 UE24EV121A_8d24d5ef-6f60-11ef-b7ef-02bdb07e3b51_20240918140313(1).ppt
x
-a---- 06-04-2025 09:47 17137664 UE24EV121A_8d24d5ef-6f60-11ef-b7ef-02bdb07e3b51_20240918140313.pptx
-a---- 26-04-2026 14:43 162282 UE24MA241B_Problems for Unit 4 - Google Docs - Unit 4 problem.pdf
-a---- 23-11-2025 10:33 301777 UE24MA242A_b1c84fa2-451e-4559-a14b-7414a554181b_20251111091359.pdf
-a---- 03-06-2025 18:54 674752 UE24ME141A_411a088a-6f5b-11ef-b7ef-02bdb07e3b51_20241231111500.pdf
-a---- 26-04-2026 11:30 1113677 Unit 3 Problem set.pdf
-a---- 27-04-2026 13:25 9523552 UNIT-3-20260427T075512Z-3-001.zip
-a---- 27-04-2026 13:25 22442442 UNIT-4-20260427T075520Z-3-001.zip
-a---- 01-05-2026 16:21 33937114 Unit3-20260501T105032Z-3-001.zip
-a---- 01-05-2026 16:21 8422617 Unit4-20260501T105038Z-3-001.zip
-a---- 16-01-2025 17:49 4410735 University of Minnesota._Downey, Allen B - Think Python_ How to
Think Like a Computer Scientist (2016_2015, O'Reilly Media) -
libgen.li.pdf
-a---- 27-10-2024 10:49 337 Untitled - Copy.ipynb
-a---- 14-10-2025 09:40 198912 Untitled 1.odg
-a---- 04-05-2025 12:48 510645 Untitled 2.pdf
-a---- 24-08-2026 14:30 123968 Untitled Diagram.drawio
-a---- 24-08-2026 15:05 114024 Untitled Diagram.drawio.pdf
-a---- 04-03-2025 20:27 133285 Untitled document.pdf
-a---- 27-10-2024 10:49 337 Untitled.ipynb
-a---- 08-11-2024 20:50 292227 Untitled3.html
-a---- 10-09-2026 14:10 260113 Untitled7.ipynb
-a---- 02-04-2025 09:17 7100 Updated_Orange_v1.xlsx
-a---- 30-09-2026 11:20 352549 Updates.pdf
-a---- 27-08-2026 12:12 277789 use-case-diagram.png
-a---- 27-08-2026 12:13 3317 use-case-flow-submit-grant-proposal.md
-a---- 07-03-2025 22:26 90675 US_Trip_ Delta Air Lines.pdf
-a---- 18-02-2025 19:08 144 utol.c
-a---- 20-04-2026 16:45 58525993 videoduke(1).dmg
-a---- 20-04-2026 16:45 58525993 videoduke.dmg
-a---- 13-04-2025 11:18 123016232 VirtualBox-7.1.6-167084-Win.exe
-a---- 18-02-2025 19:08 308 volume_cylinder.c
-a---- 22-04-2026 13:04 487894 VortexVector OS-Jackfruit main screenshots.zip
-a---- 03-05-2025 16:04 107547552 VSCodeUserSetup-x64-1.99.3.exe
-a---- 28-10-2024 17:28 143320 week 5 lab assignment.pdf
-a---- 31-08-2026 13:29 409905 Week 5_Level 1_StudentCopy.pdf
-a---- 10-09-2026 13:29 161530 Week 6 Lab Instructions.pdf
-a---- 03-09-2026 13:31 10295 Week5_Lab_Boilerplate.ipynb
-a---- 03-09-2026 13:40 13356 Week5_Lab_Boilerplate_FINISHED.ipynb
-a---- 03-09-2026 14:03 229079 Week5_Lab_Boilerplate_FINISHED_WITH_THEORY.ipynb
```

-a---- 03-09-2026 13:31 488796 Week5_Linear_vs_Logistic_Regression.pdf
-a---- 09-11-2024 09:23 177826 week6 Lab Submission Template..pdf
-a---- 10-09-2026 16:59 235585 Week6_ANN_Lab_Report.pdf
-a---- 10-09-2026 13:29 64469 Week6_Boilerplate.ipynb
-a---- 10-09-2026 13:34 49626749 Week6_Question_Walkthrough.mp4
-a---- 10-09-2026 14:26 448626 Week_6_Lab_Submission_Rohan_Chukkapalli.pdf
----- 22-04-2026 11:42 23327 WhatsApp Image 2026-04-22 at 11.38.03 AM.jpeg
-a---- 22-04-2026 11:46 476947 WhatsApp Image 2026-04-22 at 11.38.03 AM.zip
----- 22-04-2026 11:42 67113 WhatsApp Image 2026-04-22 at 11.38.05 AM.jpeg
----- 22-04-2026 11:42 15787 WhatsApp Image 2026-04-22 at 11.38.06 AM.jpeg
----- 22-04-2026 11:42 10919 WhatsApp Image 2026-04-22 at 11.38.08 AM.jpeg
----- 22-04-2026 11:42 9941 WhatsApp Image 2026-04-22 at 11.38.09 AM.jpeg
----- 22-04-2026 11:42 95437 WhatsApp Image 2026-04-22 at 11.38.11 AM.jpeg
----- 22-04-2026 11:43 131255 WhatsApp Image 2026-04-22 at 11.38.32 AM.jpeg
----- 22-04-2026 11:43 133071 WhatsApp Image 2026-04-22 at 11.38.34 AM.jpeg
-a---- 18-09-2026 14:16 251055 WhatsApp Image 2026-09-18 at 2.15.16 PM.jpeg
-a---- 27-10-2024 10:56 1058336 WhatsApp Installer.exe
-a---- 09-01-2026 15:15 96624568 Wireshark-4.6.2-x64.exe
-a---- 06-02-2026 19:26 56693 wireshark_nonpersistent_10imgs_with_images.png
-a---- 07-12-2024 13:34 82366 _week 9 Lab Submission Template.pdf

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:56:15+05:30

Completed At: 2026-09-30T11:56:26+05:30

The command exited with code 0.

Output:

Directory: C:\Users\rchuk\Downloads

Mode LastWriteTime Length Name

d----- 08-09-2026 15:33 Lab2

d----- 27-02-2025 19:19 _temp_matlab_R2024b_Windows

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:56:29+05:30

Completed At: 2026-09-30T11:56:36+05:30

The command exited with code 0.

Output:

diff.astextplain.textconv=astextplain

filter.lfs.clean=git-lfs clean -- %f

filter.lfs.smudge=git-lfs smudge -- %f

filter.lfs.process=git-lfs filter-process

filter.lfs.required=true

http.sslbackend=openssl

http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt

core.autocrlf=true

core.fscache=true

core.symlinks=false

pull.rebase=false

credential.helper=manager

credential.https://dev.azure.com.usehttppath=true

```
init.defaultbranch=master
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=VortexVector
user.email=rchukky22@gmail.com
```

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:56:42+05:30
Completed At: 2026-09-30T11:56:52+05:30

The command exited with code 1.

Output:

gh : The term 'gh' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

At line:1 char:1

+ gh --version

+ ~~

+ CategoryInfo : ObjectNotFound: (gh:String) [], CommandNotFoundException

+ FullyQualifiedErrorId : CommandNotFoundException

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:56:59+05:30
Completed At: 2026-09-30T11:57:12+05:30

The command exited with code 1.

Output:

Python 3.14.7

Traceback (most recent call last):

File "<string>", line 1, in <module>

import pygame; print(pygame.__version__)

~~~~~

ModuleNotFoundError: No module named 'pygame'

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:57:22+05:30  
Completed At: 2026-09-30T11:57:30+05:30

The command exited with code 0.

Output:

Cloning into 'c:\Users\rchuk\Downloads\09\_tic\_tac\_toe'...

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:57:38+05:30  
Completed At: 2026-09-30T11:57:43+05:30

The command exited with code 0.

Output:

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe

Mode LastWriteTime Length Name



```
-----
d---- 30-09-2026 11:57 tic-tac-toe
-a---- 30-09-2026 11:57 34 .gitignore
-a---- 30-09-2026 11:57 4641 README.md
```

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\tic-tac-toe

```
Mode LastWriteTime Length Name
-----
d---- 30-09-2026 11:57 game
-a---- 30-09-2026 11:57 967 main.py
-a---- 30-09-2026 11:57 15 requirements.txt
```

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\tic-tac-toe\game

```
Mode LastWriteTime Length Name
-----
-a---- 30-09-2026 11:57 488 ai.py
-a---- 30-09-2026 11:57 2930 game_engine.py
-a---- 30-09-2026 11:57 1936 renderer.py
-a---- 30-09-2026 11:57 620 rules.py
-a---- 30-09-2026 11:57 0 __init__.py
```

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:57:51+05:30

Completed At: 2026-09-30T11:57:51+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/main.py`

Total Lines: 43

Total Bytes: 967

Showing lines 1 to 43

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: Tic-Tac-Toe (Lab Starter)
3:
4: Run with: python3 main.py
5:
6: Click a cell to place your symbol.
7: """
8:
9: import pygame
10:
11: from game.game_engine import GameEngine
12: from game.renderer import WINDOW_SIZE
13:
14:
15: def main():
16:     pygame.init()
17:     screen = pygame.display.set_mode(WINDOW_SIZE)
18:     pygame.display.set_caption("Tic-Tac-Toe")
19:     clock = pygame.time.Clock()
20:     font = pygame.font.SysFont("consolas", 22)
21:
```

```

22: engine = GameEngine()
23: running = True
24: while running:
25:     for event in pygame.event.get():
26:         if event.type == pygame.QUIT:
27:             running = False
28:         elif event.type == pygame.MOUSEBUTTONDOWN:
29:             engine.handle_click(event.pos)
30:         elif event.type == pygame.KEYDOWN:
31:             engine.handle_keydown(event.key)
32:
33:     engine.draw(screen, font)
34:
35:     pygame.display.flip()
36:     clock.tick(60)
37:
38:     pygame.quit()
39:
40:
41: if __name__ == "__main__":
42:     main()
43:

```

The above content shows the entire, complete file contents of the requested file.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:57:59+05:30

Completed At: 2026-09-30T11:57:59+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py`

Total Lines: 78

Total Bytes: 2930

Showing lines 1 to 78

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```

1: """
2: GameEngine: owns the board, turn state, and round-end logic.
3:
4: You (the player) always play X and click to move. The computer always
5: plays O and moves automatically right after you, using a simple
6: random-move AI (see game/ai.py) - this is given infrastructure, not
7: something you need to build.
8:
9: Starter version: no scoreboard yet, no first-player choice, and only
10: one combined reset control. Win/draw detection has known bugs (see
11: game/rules.py and check_round_end below) that Task 1 asks you to fix,
12: and move validation has a known gap (see handle_click) that Task 3
13: asks you to fix.
14: """
15:
16: from game.rules import check_winner, is_board_full
17: from game.renderer import board_pos_to_cell
18: from game.ai import choose_move
19:
20: HUMAN_SYMBOL = 'X'
21: COMPUTER_SYMBOL = 'O'
22:
23:
24: class GameEngine:

```

```

25: def __init__(self):
26: self.board = [[None] * 3 for _ in range(3)]
27: self.current_player = 'X'
28: self.round_over = False
29: self.winner = None # 'X', 'O', or None (meaning draw, only valid when round_over)
30:
31: def handle_click(self, pos):
32: if self.current_player != HUMAN_SYMBOL:
33: return # not your turn - the computer is about to move (or already has)
34: cell = board_pos_to_cell(pos)
35: if cell is None:
36: return
37: row, col = cell
38: self.board[row][col] = self.current_player # BUG: doesn't check if the cell is already occupied
39: self.check_round_end()
40: self.current_player = 'O' if self.current_player == 'X' else 'X'
41: self._maybe_take_computer_turn()
42:
43: def _maybe_take_computer_turn(self):
44: if self.round_over or self.current_player != COMPUTER_SYMBOL:
45: return
46: move = choose_move(self.board)
47: if move is None:
48: return
49: row, col = move
50: self.board[row][col] = self.current_player
51: self.check_round_end()
52: self.current_player = 'O' if self.current_player == 'X' else 'X'
53:
54: def handle_keydown(self, key):
55: import pygame
56: if key == pygame.K_r:
57: self.__init__()
58:
59: def check_round_end(self):
60: if is_board_full(self.board): # BUG: checked before looking for a winner
61: self.round_over = True
62: self.winner = None
63: return
64: winner = check_winner(self.board)
65: if winner:
66: self.round_over = True
67: self.winner = winner
68:
69: def draw(self, surface, font):
70: from game import renderer
71: renderer.draw_board(surface, self.board)
72: turn_label = "Your turn (X)" if self.current_player == HUMAN_SYMBOL else "Computer's turn (O)"
73: renderer.draw_text(surface, font, turn_label, (10, 20))
74:
75: if self.round_over:
76: text = f"{self.winner} wins!" if self.winner else "Draw!"
77: renderer.draw_banner(surface, font, f"{text} Press R for a new round.")
78:

```

The above content shows the entire, complete file contents of the requested file.

Created At: 2026-09-30T11:58:08+05:30

Completed At: 2026-09-30T11:58:09+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/rules.py`

Total Lines: 24

Total Bytes: 620

Showing lines 1 to 24

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: rules: win/draw detection for a 3x3 board.
3:
4: board is a 3x3 list of lists, each cell is None, 'X', or 'O'.
5: """
6:
7:
8: def check_winner(board):
9:     lines = []
10:    for r in range(3):
11:        lines.append([board[r][0], board[r][1], board[r][2]])
12:    for c in range(3):
13:        lines.append([board[0][c], board[1][c], board[2][c]])
14:    # NOTE: diagonals are not included here
15:
16:    for line in lines:
17:        if line[0] is not None and line[0] == line[1] == line[2]:
18:            return line[0]
19:    return None
20:
21:
22: def is_board_full(board):
23:    return all(cell is not None for row in board for cell in row)
24:
```

The above content shows the entire, complete file contents of the requested file.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:58:17+05:30

Completed At: 2026-09-30T11:58:17+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/ai.py`

Total Lines: 19

Total Bytes: 488

Showing lines 1 to 19

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: ai: a simple computer opponent for the second player (O).
3:
4: Picks a random empty cell - this is intentionally simple. The point
5: of this lab is testing your own win/draw/turn logic against a working
6: opponent, not building a hard-to-beat AI.
7: """
8:
9: import random
10:
11:
12: def choose_move(board):
13:     empty_cells = [
14:         (r, c) for r in range(3) for c in range(3) if board[r][c] is None
15:     ]
```

```
16: if not empty_cells:
17: return None
18: return random.choice(empty_cells)
19:
```

The above content shows the entire, complete file contents of the requested file.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:58:32+05:30

Completed At: 2026-09-30T11:58:32+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/renderer.py`

Total Lines: 58

Total Bytes: 1936

Showing lines 1 to 58

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: renderer: all pygame drawing lives here, kept separate from game logic.
3: """
4:
5: import pygame
6:
7: WIDTH, HEIGHT = 400, 500
8: BOARD_SIZE = 360
9: CELL_SIZE = BOARD_SIZE // 3
10: BOARD_TOP = 100
11: WINDOW_SIZE = (WIDTH, HEIGHT)
12:
13: COLOR_BG = (245, 245, 245)
14: COLOR_LINE = (60, 60, 60)
15: COLOR_X = (200, 60, 60)
16: COLOR_O = (60, 100, 200)
17: COLOR_TEXT = (30, 30, 30)
18:
19:
20: def board_pos_to_cell(pos):
21: x, y = pos
22: y -= BOARD_TOP
23: if not (0 <= x < BOARD_SIZE and 0 <= y < BOARD_SIZE):
24: return None
25: col = x // CELL_SIZE
26: row = y // CELL_SIZE
27: return int(row), int(col)
28:
29:
30: def draw_board(surface, board):
31: surface.fill(COLOR_BG)
32: for i in range(1, 3):
33: pygame.draw.line(surface, COLOR_LINE, (i * CELL_SIZE, BOARD_TOP), (i * CELL_SIZE, BOARD_TOP + BOARD_SIZE), 3)
34: pygame.draw.line(surface, COLOR_LINE, (0, BOARD_TOP + i * CELL_SIZE), (BOARD_SIZE, BOARD_TOP + i * CELL_SIZE), 3)
35:
36: for r in range(3):
37: for c in range(3):
38: symbol = board[r][c]
39: if symbol is None:
40: continue
41: center = (c * CELL_SIZE + CELL_SIZE // 2, BOARD_TOP + r * CELL_SIZE + CELL_SIZE // 2)
42: if symbol == 'X':
```

```
43: offset = CELL_SIZE // 3
44: pygame.draw.line(surface, COLOR_X, (center[0]-offset, center[1]-offset), (center[0]+offset, center[1]+offset), 6)
45: pygame.draw.line(surface, COLOR_X, (center[0]+offset, center[1]-offset), (center[0]-offset, center[1]+offset), 6)
46: else:
47: pygame.draw.circle(surface, COLOR_O, center, CELL_SIZE // 3, 6)
48:
49:
50: def draw_text(surface, font, text, pos, color=COLOR_TEXT):
51: surface.blit(font.render(text, True, color), pos)
52:
53:
54: def draw_banner(surface, font, text):
55: surf = font.render(text, True, (180, 40, 40))
56: rect = surf.get_rect(center=(surface.get_width() // 2, BOARD_TOP + BOARD_SIZE + 40))
57: surface.blit(surf, rect)
58:
```

The above content shows the entire, complete file contents of the requested file.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:58:45+05:30

Tool is running as a background task with task id: 797e1d1a-07d3-4d1b-a401-45de60e4324a/task-34

Task Description: pip install pygame

Task logs are available at:

file:///C:/Users/rchuk/.gemini/antigravity/brain/797e1d1a-07d3-4d1b-a401-45de60e4324a/.system\_generated/tasks/task-34.log

YOU MUST TAKE ONE OF THE FOLLOWING TWO ACTIONS: A) either proceed to other relevant work (if any) or, B) simply update the user with a short message (that you have launched the command and will wait for it to finish) and end the turn.

DO NOTHING ELSE.

---

### Antigravity AI (Pair Programmer)

I have started installing `pygame` in the background and will continue once the installation finishes.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:59:38+05:30

Completed At: 2026-09-30T11:59:43+05:30

The command exited with code 0.

Output:

-V:3.14 \* Python 3.14 (64-bit)

-V:3.13 Python 3.13 (64-bit)

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T11:59:47+05:30

Tool is running as a background task with task id: 797e1d1a-07d3-4d1b-a401-45de60e4324a/task-40

Task Description: py -3.13 -m pip install pygame

Task logs are available at:

file:///C:/Users/rchuk/.gemini/antigravity/brain/797e1d1a-07d3-4d1b-a401-45de60e4324a/.system\_generated/tasks/task-40.log

YOU MUST TAKE ONE OF THE FOLLOWING TWO ACTIONS: A) either proceed to other relevant work (if any) or, B) simply update the user with a short message (that you have launched the command and will wait for it to finish) and end the turn.

DO NOTHING ELSE.

---

### Antigravity AI (Pair Programmer)

I'm installing `pygame` on Python 3.13 in the background to ensure the game can run smoothly.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:00:17+05:30  
Completed At: 2026-09-30T12:00:26+05:30

The command exited with code 0.

Output:

pygame 2.6.1 (SDL 2.28.4, Python 3.13.0)  
Hello from the pygame community. <https://www.pygame.org/contribute.html>  
2.6.1

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:00:33+05:30  
Completed At: 2026-09-30T12:00:40+05:30

The command exited with code 1.

Output:

fatal: not a git repository (or any of the parent directories): .git

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:00:50+05:30  
Completed At: 2026-09-30T12:00:56+05:30

The command exited with code 0.

Output:

Directory: C:\Users\rchuk\Downloads\Lab2

Mode LastWriteTime Length Name

-----

-a---- 08-09-2026 15:24 908987 PES1UG24CS135\_ROHAN\_Lab\_2\_Jira\_Submission.pdf

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:01:04+05:30  
Completed At: 2026-09-30T12:01:09+05:30

The command exited with code 0.

Stdout:

Stderr:

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:01:19+05:30  
Completed At: 2026-09-30T12:01:23+05:30

The command exited with code 0.

Output:

On branch main

Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:01:27+05:30  
Completed At: 2026-09-30T12:01:47+05:30

The command exited with code 0.

Stdout:

Stderr:

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:01:52+05:30

Completed At: 2026-09-30T12:01:52+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/.gitignore`

Total Lines: 4

Total Bytes: 34

Showing lines 1 to 4

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

1: \_\_pycache\_\_/

2: \*.pyc

3: .venv/

4: venv/

The above content shows the entire, complete file contents of the requested file.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:01:55+05:30

Completed At: 2026-09-30T12:01:56+05:30

Encountered error in step execution: Could not successfully apply any edits. Please review the file and try smaller edits that you are more confident about.: chunk 0: target content not found in file

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:01:59+05:30

Completed At: 2026-09-30T12:02:06+05:30

The command exited with code 0.

Output:

\_\_pycache\_\_/

\*.pyc

.venv/

venv/

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:02:11+05:30

Completed At: 2026-09-30T12:02:15+05:30

The command exited with code 0.

Stdout:

Stderr:

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:02:34+05:30

Completed At: 2026-09-30T12:02:34+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/rules.py`

Total Lines: 24

Total Bytes: 620

Showing lines 1 to 24



The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: rules: win/draw detection for a 3x3 board.
3:
4: board is a 3x3 list of lists, each cell is None, 'X', or 'O'.
5: """
6:
7:
8: def check_winner(board):
9:     lines = []
10:    for r in range(3):
11:        lines.append([board[r][0], board[r][1], board[r][2]])
12:    for c in range(3):
13:        lines.append([board[0][c], board[1][c], board[2][c]])
14:    # NOTE: diagonals are not included here
15:
16:    for line in lines:
17:        if line[0] is not None and line[0] == line[1] == line[2]:
18:            return line[0]
19:    return None
20:
21:
22: def is_board_full(board):
23:    return all(cell is not None for row in board for cell in row)
24:
```

The above content shows the entire, complete file contents of the requested file.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:02:39+05:30

Completed At: 2026-09-30T12:02:39+05:30

The following changes were made by the replace\_file\_content tool to: c:\Users\rchuk\Downloads\09\_tic\_tac\_toe\tic-tac-toe\game\rules.py. If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

[diff\_block\_start]

@@ -11,7 +11,9 @@

lines.append([board[r][0], board[r][1], board[r][2]])

for c in range(3):

lines.append([board[0][c], board[1][c], board[2][c]])

- # NOTE: diagonals are not included here

+ # Include both diagonals

+ lines.append([board[0][0], board[1][1], board[2][2]])

+ lines.append([board[0][2], board[1][1], board[2][0]])

for line in lines:

if line[0] is not None and line[0] == line[1] == line[2]:

[diff\_block\_end]

Please note that the above snippet only shows the MODIFIED lines from the last change. It shows up to 3 lines of unchanged lines before and after the modified lines. The actual file contents may have many more lines not shown.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:02:46+05:30

Completed At: 2026-09-30T12:02:46+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py`

Total Lines: 78

Total Bytes: 2930

Showing lines 1 to 78

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: GameEngine: owns the board, turn state, and round-end logic.
3:
4: You (the player) always play X and click to move. The computer always
5: plays O and moves automatically right after you, using a simple
6: random-move AI (see game/ai.py) - this is given infrastructure, not
7: something you need to build.
8:
9: Starter version: no scoreboard yet, no first-player choice, and only
10: one combined reset control. Win/draw detection has known bugs (see
11: game/rules.py and check_round_end below) that Task 1 asks you to fix,
12: and move validation has a known gap (see handle_click) that Task 3
13: asks you to fix.
14: """
15:
16: from game.rules import check_winner, is_board_full
17: from game.renderer import board_pos_to_cell
18: from game.ai import choose_move
19:
20: HUMAN_SYMBOL = 'X'
21: COMPUTER_SYMBOL = 'O'
22:
23:
24: class GameEngine:
25:     def __init__(self):
26:         self.board = [[None] * 3 for _ in range(3)]
27:         self.current_player = 'X'
28:         self.round_over = False
29:         self.winner = None # 'X', 'O', or None (meaning draw, only valid when round_over)
30:
31:     def handle_click(self, pos):
32:         if self.current_player != HUMAN_SYMBOL:
33:             return # not your turn - the computer is about to move (or already has)
34:         cell = board_pos_to_cell(pos)
35:         if cell is None:
36:             return
37:         row, col = cell
38:         self.board[row][col] = self.current_player # BUG: doesn't check if the cell is already occupied
39:         self.check_round_end()
40:         self.current_player = 'O' if self.current_player == 'X' else 'X'
41:         self._maybe_take_computer_turn()
42:
43:     def _maybe_take_computer_turn(self):
44:         if self.round_over or self.current_player != COMPUTER_SYMBOL:
45:             return
46:         move = choose_move(self.board)
47:         if move is None:
48:             return
49:         row, col = move
50:         self.board[row][col] = self.current_player
51:         self.check_round_end()
52:         self.current_player = 'O' if self.current_player == 'X' else 'X'
53:
54:     def handle_keydown(self, key):
55:         import pygame
```

```

56: if key == pygame.K_r:
57: self.__init__()
58:
59: def check_round_end(self):
60: if is_board_full(self.board): # BUG: checked before looking for a winner
61: self.round_over = True
62: self.winner = None
63: return
64: winner = check_winner(self.board)
65: if winner:
66: self.round_over = True
67: self.winner = winner
68:
69: def draw(self, surface, font):
70: from game import renderer
71: renderer.draw_board(surface, self.board)
72: turn_label = "Your turn (X)" if self.current_player == HUMAN_SYMBOL else "Computer's turn (O)"
73: renderer.draw_text(surface, font, turn_label, (10, 20))
74:
75: if self.round_over:
76: text = f"{self.winner} wins!" if self.winner else "Draw!"
77: renderer.draw_banner(surface, font, f"{text} Press R for a new round.")
78:

```

The above content shows the entire, complete file contents of the requested file.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:02:53+05:30

Completed At: 2026-09-30T12:02:53+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/renderer.py`

Total Lines: 58

Total Bytes: 1936

Showing lines 1 to 58

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```

1: """
2: renderer: all pygame drawing lives here, kept separate from game logic.
3: """
4:
5: import pygame
6:
7: WIDTH, HEIGHT = 400, 500
8: BOARD_SIZE = 360
9: CELL_SIZE = BOARD_SIZE // 3
10: BOARD_TOP = 100
11: WINDOW_SIZE = (WIDTH, HEIGHT)
12:
13: COLOR_BG = (245, 245, 245)
14: COLOR_LINE = (60, 60, 60)
15: COLOR_X = (200, 60, 60)
16: COLOR_O = (60, 100, 200)
17: COLOR_TEXT = (30, 30, 30)
18:
19:
20: def board_pos_to_cell(pos):
21: x, y = pos
22: y -= BOARD_TOP
23: if not (0 <= x < BOARD_SIZE and 0 <= y < BOARD_SIZE):

```

```

24: return None
25: col = x // CELL_SIZE
26: row = y // CELL_SIZE
27: return int(row), int(col)
28:
29:
30: def draw_board(surface, board):
31: surface.fill(COLOR_BG)
32: for i in range(1, 3):
33: pygame.draw.line(surface, COLOR_LINE, (i * CELL_SIZE, BOARD_TOP), (i * CELL_SIZE, BOARD_TOP + BOARD_SIZE), 3)
34: pygame.draw.line(surface, COLOR_LINE, (0, BOARD_TOP + i * CELL_SIZE), (BOARD_SIZE, BOARD_TOP + i * CELL_SIZE), 3)
35:
36: for r in range(3):
37: for c in range(3):
38: symbol = board[r][c]
39: if symbol is None:
40: continue
41: center = (c * CELL_SIZE + CELL_SIZE // 2, BOARD_TOP + r * CELL_SIZE + CELL_SIZE // 2)
42: if symbol == 'X':
43: offset = CELL_SIZE // 3
44: pygame.draw.line(surface, COLOR_X, (center[0]-offset, center[1]-offset), (center[0]+offset, center[1]+offset), 6)
45: pygame.draw.line(surface, COLOR_X, (center[0]+offset, center[1]-offset), (center[0]-offset, center[1]+offset), 6)
46: else:
47: pygame.draw.circle(surface, COLOR_O, center, CELL_SIZE // 3, 6)
48:
49:
50: def draw_text(surface, font, text, pos, color=COLOR_TEXT):
51: surface.blit(font.render(text, True, color), pos)
52:
53:
54: def draw_banner(surface, font, text):
55: surf = font.render(text, True, (180, 40, 40))
56: rect = surf.get_rect(center=(surface.get_width() // 2, BOARD_TOP + BOARD_SIZE + 40))
57: surface.blit(surf, rect)
58:

```

The above content shows the entire, complete file contents of the requested file.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:03:11+05:30

Completed At: 2026-09-30T12:03:11+05:30

Created file file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py with requested content.

If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:03:21+05:30

Completed At: 2026-09-30T12:03:22+05:30

Created file file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/renderer.py with requested content.

If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:03:28+05:30

Completed At: 2026-09-30T12:03:29+05:30

File Path: `file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/main.py`

Total Lines: 43

Total Bytes: 967

Showing lines 1 to 43

The following code has been modified to include a line number before every line, in the format: <line\_number>: <original\_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.

```
1: """
2: Tic-Tac-Toe (Lab Starter)
3:
4: Run with: python3 main.py
5:
6: Click a cell to place your symbol.
7: """
8:
9: import pygame
10:
11: from game.game_engine import GameEngine
12: from game.renderer import WINDOW_SIZE
13:
14:
15: def main():
16:     pygame.init()
17:     screen = pygame.display.set_mode(WINDOW_SIZE)
18:     pygame.display.set_caption("Tic-Tac-Toe")
19:     clock = pygame.time.Clock()
20:     font = pygame.font.SysFont("consolas", 22)
21:
22:     engine = GameEngine()
23:     running = True
24:     while running:
25:         for event in pygame.event.get():
26:             if event.type == pygame.QUIT:
27:                 running = False
28:             elif event.type == pygame.MOUSEBUTTONDOWN:
29:                 engine.handle_click(event.pos)
30:             elif event.type == pygame.KEYDOWN:
31:                 engine.handle_keydown(event.key)
32:
33:     engine.draw(screen, font)
34:
35:     pygame.display.flip()
36:     clock.tick(60)
37:
38:     pygame.quit()
39:
40:
41: if __name__ == "__main__":
42:     main()
43:
```

The above content shows the entire, complete file contents of the requested file.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:03:43+05:30

Completed At: 2026-09-30T12:03:44+05:30

Created file file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/test\_game.py with requested content.

If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:03:47+05:30  
Completed At: 2026-09-30T12:04:04+05:30

The command exited with code 0.

Output:

.....

Ran 7 tests in 0.001s

OK

pygame 2.6.1 (SDL 2.28.4, Python 3.13.0)

Hello from the pygame community. <https://www.pygame.org/contribute.html>

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:04:10+05:30

Completed At: 2026-09-30T12:04:11+05:30

The command exited with code 0.

Output:

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: .gitignore

modified: tic-tac-toe/game/game\_engine.py

modified: tic-tac-toe/game/renderer.py

modified: tic-tac-toe/game/rules.py

Untracked files:

(use "git add <file>..." to include in what will be committed)

tic-tac-toe/test\_game.py

no changes added to commit (use "git add" and/or "git commit -a")

---

## Antigravity AI (Pair Programmer)

Everything is set up, fixed, and verified with unit tests. Below is the complete walkthrough, prompt history (for your doc/PDF deliverable), video recording instructions, and git push instructions.

---

### Step-by-Step VibeCoding Prompts & Solutions Log

\*(You can export this chat history directly as your required doc/pdf deliverable)\*

##### Attempt 1: Fix Win & Draw Detection (Task 1)

- **Problem:** Diagonal wins were completely ignored; when a win happened on the 9th move (filling the board), it erroneously reported "Draw!"; once a round ended, empty cells could still be clicked.

- **Solution Applied:**

1. In [game/rules.py](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/rules.py), added both diagonals [board[0][0], board[1][1], board[2][2]] and [board[0][2], board[1][1], board[2][0]] to check\_winner().

2. In [game/game\_engine.py](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py), updated check\_round\_end() to check check\_winner() **before** is\_board\_full().

3. Added an early return check if self.round\_over: return at the top of handle\_click().

##### Attempt 2: Turn & Move Validation (Task 3)

- **Problem:** Clicking on an already-occupied cell overwrote the existing symbol.

- **Solution Applied:**

1. In `handle_click()` inside `[game/game_engine.py]`(file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py), added occupancy validation:

```
```python
if self.board[row][col] is not None:
    return
```
```

2. Clicking an occupied cell rejects the move cleanly without modifying the board or advancing the turn.

#### Attempt 3: Persistent Scoreboard & Controls (Tasks 2 & 4)

- **Requirements:**

- Track X wins, O wins, and Draws across rounds.
- Keep scores on round restart (`R`), clear scores on match reset (`M`).
- Allow selecting first player (`X` or `O`) with `T` toggle; when `O` starts, the computer takes the first turn automatically.

- **Solution Applied:**

1. In `[game/game_engine.py]`(file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py):

- Added `self.scores = {'X': 0, 'O': 0, 'Draw': 0}` and `self.starting_player = 'X'`.
- Added `new_round()` (triggered by `R`), `reset_match()` (triggered by `M`), and `toggle_starter()` (triggered by `T`).
- Automated computer move on round start if `starting_player == 'O'`.

2. In `[game/renderer.py]`(file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/renderer.py):

- Rendered persistent scoreboard: `X: 0 | O: 0 | Draws: 0`.
- Rendered current turn, starter toggle indicator, result banners, and control keys (`[R]`, `[M]`, `[T]`).

---

### Verification: Unit Test Results

A test suite `[tic-tac-toe/test_game.py]`(file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/test\_game.py) was run covering all 4 tasks:

```
```text
```

Ran 7 tests in 0.001s

OK

- test\_diagonal\_wins: PASSED (both diagonals detected)
- test\_win\_on\_last\_move\_is\_not\_draw: PASSED (winner prioritized over board-full)
- test\_genuine\_draw: PASSED (draw detected correctly)
- test\_occupied\_cell\_move\_rejected: PASSED (cannot overwrite cell)
- test\_no\_moves\_accepted\_after\_round\_over: PASSED (board locked when finished)
- test\_persistent\_scoreboard: PASSED (persists on R, clears on M)
- test\_starting\_player\_choice\_and\_computer\_first\_move: PASSED (O start triggers AI move)

```
```
```

---

### Instructions for Recording Deliverables

#### 1. "Before" Video (10 Seconds)

The untouched starter code is preserved in `tic-tac-toe-before`:

1. Open PowerShell and run:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe\tic-tac-toe-before"
py -3.13 main.py
```
```

2. Start your screen recording (`Win + Alt + R` on Windows).

3. Play a diagonal 3-in-a-row to show that no win is registered, and click an occupied cell to show it gets overwritten.

4. Stop recording and save the video (e.g., `before.mp4`).

#### 2. "After" Video (10 Seconds)

The updated code is in `tic-tac-toe`:

1. In PowerShell, run:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe\tic-tac-toe"
py -3.13 main.py
```
```

2. Start screen recording (`Win + Alt + R`).
3. Complete a diagonal win: observe the ``X wins!`` banner and the scoreboard update to `X: 1`.
4. Try clicking an already marked cell: observe that nothing gets overwritten.
5. Press `R` to start a new round (scoreboard stays intact).
6. Press `T` to toggle the starter to `O`: observe computer making the first move.
7. Press `M`: observe match reset and scoreboard returning to zero.
8. Stop recording and save the video (e.g., `after.mp4`).

---

### ### Pushing to Your Personal Repository

\*(Remember: Do not raise a PR to the main `SETAPESU26` repository)\*

In PowerShell:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe"
```

# 1. Commit your changes

```
git add .
git commit -m "Fix win/draw logic and implement scoreboard, move validation, and first-player choice"
```

# 2. Point origin to your own GitHub repository

```
# (Create a new repo on your GitHub account, e.g. '09_tic_tac_toe' or 'Lab-4')
git remote set-url origin https://github.com/VortexVector/<your-repo-name>.git
```

# 3. Push your code

```
git push -u origin main
```
```

---

## User

dont ask me always for permission always do wt is needed

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:04:35+05:30

Completed At: 2026-09-30T12:04:36+05:30

The command exited with code 0.

Output:

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: .gitignore

modified: tic-tac-toe/game/game\_engine.py

modified: tic-tac-toe/game/renderer.py

modified: tic-tac-toe/game/rules.py

Untracked files:

(use "git add <file>..." to include in what will be committed)

tic-tac-toe/test\_game.py



no changes added to commit (use "git add" and/or "git commit -a")

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:04:44+05:30

Tool is running as a background task with task id: 797e1d1a-07d3-4d1b-a401-45de60e4324a/task-88

Task Description: Get-ChildItem -Path "c:\Users\rchuk\Documents", "c:\Users\rchuk\Desktop", "c:\Users\rchuk\Downloads" -Recurse -Depth 2 -Directory -Filter ".git" -ErrorAction SilentlyContinue | Select-Object FullName

Task logs are available at:

file:///C:/Users/rchuk/.gemini/antigravity/brain/797e1d1a-07d3-4d1b-a401-45de60e4324a/.system\_generated/tasks/task-88.log

YOU MUST TAKE ONE OF THE FOLLOWING TWO ACTIONS: A) either proceed to other relevant work (if any) or, B) simply update the user with a short message (that you have launched the command and will wait for it to finish) and end the turn.

DO NOTHING ELSE.

---

### Antigravity AI (Pair Programmer)

I am searching for any existing coursework git repositories on your system to see where your previous labs are kept.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:05:18+05:30

Completed At: 2026-09-30T12:05:22+05:30

The command exited with code 0.

Output:

docx: not available

reportlab: not available

weasyprint: not available

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:05:28+05:30

Tool is running as a background task with task id: 797e1d1a-07d3-4d1b-a401-45de60e4324a/task-94

Task Description: py -3.13 -m pip install python-docx reportlab

Task logs are available at:

file:///C:/Users/rchuk/.gemini/antigravity/brain/797e1d1a-07d3-4d1b-a401-45de60e4324a/.system\_generated/tasks/task-94.log

YOU MUST TAKE ONE OF THE FOLLOWING TWO ACTIONS: A) either proceed to other relevant work (if any) or, B) simply update the user with a short message (that you have launched the command and will wait for it to finish) and end the turn.

DO NOTHING ELSE.

---

### Antigravity AI (Pair Programmer)

I am installing `reportlab` and `python-docx` to automatically generate the doc and PDF deliverables for your submission.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:06:14+05:30

Completed At: 2026-09-30T12:06:14+05:30

Created file file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/generate\_report.py with requested content.

If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:06:17+05:30

Completed At: 2026-09-30T12:06:23+05:30

The command exited with code 0.

Output:

DOCX created at: c:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4\PES1UG24CS135\_Lab4\_Chat\_History.docx

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:06:28+05:30  
Completed At: 2026-09-30T12:06:33+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4

| Mode   | LastWriteTime    | Length | Name                                 |
|--------|------------------|--------|--------------------------------------|
| ----   | -----            | -----  | ----                                 |
| -a---- | 30-09-2026 12:06 | 38535  | PES1UG24CS135_Lab4_Chat_History.docx |
| -a---- | 30-09-2026 12:06 | 4184   | PES1UG24CS135_Lab4_Chat_History.pdf  |

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:06:39+05:30  
Completed At: 2026-09-30T12:06:43+05:30

The command exited with code 0.  
Stdout:

Stderr:

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:06:49+05:30  
Completed At: 2026-09-30T12:06:49+05:30  
Created file file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4/Videos/README\_VIDEOS.txt with requested content.  
If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:06:53+05:30  
Completed At: 2026-09-30T12:06:58+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4

| Mode   | LastWriteTime    | Length | Name                                 |
|--------|------------------|--------|--------------------------------------|
| ----   | -----            | -----  | ----                                 |
| d----- | 30-09-2026 12:06 |        | Updated_Code                         |
| d----- | 30-09-2026 12:06 |        | Videos                               |
| -a---- | 30-09-2026 12:06 | 38535  | PES1UG24CS135_Lab4_Chat_History.docx |
| -a---- | 30-09-2026 12:06 | 4184   | PES1UG24CS135_Lab4_Chat_History.pdf  |

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4\Updated\_Code

Mode LastWriteTime Length Name

-----

d---- 30-09-2026 12:06 game  
-a---- 30-09-2026 11:57 967 main.py  
-a---- 30-09-2026 11:57 15 requirements.txt  
-a---- 30-09-2026 12:03 3809 test\_game.py

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4\Updated\_Code\game

Mode LastWriteTime Length Name

-----

d---- 30-09-2026 12:06 \_\_pycache\_\_  
-a---- 30-09-2026 11:57 488 ai.py  
-a---- 30-09-2026 12:03 4495 game\_engine.py  
-a---- 30-09-2026 12:03 4282 renderer.py  
-a---- 30-09-2026 12:02 723 rules.py  
-a---- 30-09-2026 11:57 0 \_\_init\_\_.py

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4\Updated\_Code\game\\_\_pycache\_\_

Mode LastWriteTime Length Name

-----

-a---- 30-09-2026 12:04 872 ai.cpython-313.pyc  
-a---- 30-09-2026 12:04 6545 game\_engine.cpython-313.pyc  
-a---- 30-09-2026 12:04 5830 renderer.cpython-313.pyc  
-a---- 30-09-2026 12:04 1557 rules.cpython-313.pyc  
-a---- 30-09-2026 12:04 165 \_\_init\_\_.cpython-313.pyc

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4\Videos

Mode LastWriteTime Length Name

-----

-a---- 30-09-2026 12:06 258 README\_VIDEOS.txt

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:07:02+05:30

Completed At: 2026-09-30T12:07:03+05:30

The command exited with code 0.

Output:

On branch main

Your branch is up to date with 'origin/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: .gitignore

modified: tic-tac-toe/game/game\_engine.py

modified: tic-tac-toe/game/renderer.py

modified: tic-tac-toe/game/rules.py

Untracked files:

(use "git add <file>..." to include in what will be committed)

Lab-4/

generate\_report.py

tic-tac-toe/test\_game.py

no changes added to commit (use "git add" and/or "git commit -a")

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:07:08+05:30

Completed At: 2026-09-30T12:07:13+05:30

The command exited with code 0.

Output:

warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'tic-tac-toe/game/game\_engine.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'tic-tac-toe/game/renderer.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.pdf', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'Lab-4/Updated\_Code/game/game\_engine.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'Lab-4/Updated\_Code/game/renderer.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'Lab-4/Updated\_Code/test\_game.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'Lab-4/Videos/README\_VIDEOS.txt', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'generate\_report.py', LF will be replaced by CRLF the next time Git touches it

warning: in the working copy of 'tic-tac-toe/test\_game.py', LF will be replaced by CRLF the next time Git touches it

[main 6460ef4] Complete Lab 4: fix win/draw detection, move validation, persistent scoreboard, and starter choice with deliverables

17 files changed, 1055 insertions(+), 41 deletions(-)

create mode 100644 Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.docx

create mode 100644 Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.pdf

create mode 100644 Lab-4/Updated\_Code/game/\_\_init\_\_.py

create mode 100644 Lab-4/Updated\_Code/game/ai.py

create mode 100644 Lab-4/Updated\_Code/game/game\_engine.py

create mode 100644 Lab-4/Updated\_Code/game/renderer.py

create mode 100644 Lab-4/Updated\_Code/game/rules.py

create mode 100644 Lab-4/Updated\_Code/main.py

create mode 100644 Lab-4/Updated\_Code/requirements.txt

create mode 100644 Lab-4/Updated\_Code/test\_game.py

create mode 100644 Lab-4/Videos/README\_VIDEOS.txt

create mode 100644 generate\_report.py

create mode 100644 tic-tac-toe/test\_game.py

---

## Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:07:20+05:30

Completed At: 2026-09-30T12:07:25+05:30

The command exited with code 0.

Output:

commit 6460ef48a990095487d6c099df22bc88722565f7

Author: VortexVector <rchukky22@gmail.com>

Date: Wed Sep 30 12:07:13 2026 +0530

Complete Lab 4: fix win/draw detection, move validation, persistent scoreboard, and starter choice with deliverables

commit 28263985b9846c3f5a716dd971f91ee752223a04

Author: Anusha Balamurali <anushaba0@gmail.com>

Date: Thu Sep 24 21:18:54 2026 +0530

## Antigravity AI (Pair Programmer)

All tasks and deliverables have been implemented, tested, and committed.

### ### What Was Done

#### 1. \*\*Fixed Broken Starter Code\*\*:

- \*\*Win Detection\*\*: Added both main diagonal and anti-diagonal checks in [game/rules.py](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/rules.py).
- \*\*Win vs. Draw Priority\*\*: Reordered `check\_round\_end()` in [game/game\_engine.py](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/game\_engine.py) to check for a winner \*before\* checking board fullness (a win on the 9th move is now correctly awarded).
- \*\*Post-Game Move Locking\*\*: In `handle\_click()`, added an early exit guard `if self.round\_over: return` to lock the board after round completion.
- \*\*Move Validation\*\*: Added `if self.board[row][col] is not None: return` in `handle\_click()` so occupied cells cannot be overwritten.

#### 2. \*\*Added All Required Features\*\*:

- \*\*Persistent Scoreboard\*\*: Integrated `self.scores` (`X`, `O`, `Draw`), updating dynamically and rendered cleanly on screen via [game/render.py](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/game/render.py).
- \*\*Dual Restart Controls\*\*:
  - `[R]`: Starts a new round, resetting the board while preserving scores.
  - `[M]`: Resets the entire match, clearing scores back to zero.
- \*\*First-Player Choice\*\*:
  - `[T]`: Toggles starter between `X` and `O`. When `O` is chosen to start, the AI automatically plays the first move.

#### 3. \*\*Automated Unit Testing\*\*:

- Built and ran a 7-test suite ([test\_game.py](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/tic-tac-toe/test\_game.py)) verifying all 4 tasks:  
```text  
Ran 7 tests in 0.001s — OK (All tests passed)  
```

#### 4. \*\*Lab-4 Deliverables Generated\*\*:

- Located at: [c:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4)
- `PES1UG24CS135\_Lab4\_Chat\_History.docx` (Chat history deliverable in doc format)
- `PES1UG24CS135\_Lab4\_Chat\_History.pdf` (Chat history deliverable in PDF format)
- `Updated\_Code/` (Full copy of the updated game and tests)
- `Videos/` (Folder ready for your before/after video clips)

#### 5. \*\*Git Commit Completed\*\*:

- Committed to `main` with hash `6460ef4`.

---

### ### Commands to Run & Record

#### #### 1. Record "Before" Video (10s)

Using the untouched backup starter code:

```
```powershell  
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe\tic-tac-toe-before"  
py -3.13 main.py  
```
```

\*(Demonstrate: diagonal 3-in-a-row not registering, clicking an occupied cell overwriting it).\*

#### #### 2. Record "After" Video (10s)

Using the updated codebase:

```
```powershell
```

```
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe\tic-tac-toe"
py -3.13 main.py
---
```

\*(Demonstrate: diagonal win recognized, scoreboard updating, occupied cell click rejected, `R` to restart round, `T` to toggle starter to O, `M` to reset match).\*

Save both `.mp4` recordings directly into

```
[c:\Users\rchuk\Downloads\09_tic_tac_toe\Lab-4\Videos](file:///c:/Users/rchuk/Downloads/09_tic_tac_toe/Lab-4/Videos).
```

---

### Push to Your Personal GitHub

Create a new repo on your GitHub account (`VortexVector`), then run:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe"
git remote set-url origin https://github.com/VortexVector/<your-repo-name>.git
git push -u origin main
```
```

---

## User

is it done

---

## Antigravity AI (Pair Programmer)

**\*\*Yes, all coding, bug fixing, feature additions, test verifications, and document generation are 100% done and committed.\*\***

### What is complete:

- \*\*Task 1 (Win/Draw Fix):\*\*** Diagonal wins detect correctly, full-board win registers as win (not draw), board locks when round ends.
- \*\*Task 2 (Scoreboard):\*\*** Real-time scoreboard persists across rounds and resets only on match reset.
- \*\*Task 3 (Move Validation):\*\*** Clicks on occupied cells are rejected without altering the board or turn.
- \*\*Task 4 (Starter Choice & Dual Controls):\*\*** `T` toggles starter (`O` triggers AI first move), `R` restarts round, `M` resets match.
- \*\*Unit Tests:\*\*** All 7 automated unit tests passed cleanly (`test\_game.py`).
- \*\*Deliverables Created:\*\***
  - [Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.pdf](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.pdf)
  - [Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.docx](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.docx)
  - [Lab-4/Updated\_Code/](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4/Updated\_Code)
- \*\*Git:\*\*** Changes committed to local git repository.

---

### Only two quick steps left for you:

- \*\*Record the two 10-second videos (`Win + Alt + R`):\*\***
  - **\*\*Before video:\*\*** run `cd "c:\Users\rchuk\Downloads\09\_tic\_tac\_toe\tic-tac-toe-before"; py -3.13 main.py`
  - **\*\*After video:\*\*** run `cd "c:\Users\rchuk\Downloads\09\_tic\_tac\_toe\tic-tac-toe"; py -3.13 main.py`
  - Save both into [Lab-4/Videos/](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4/Videos).
- \*\*Push to your GitHub repository:\*\***

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe"
git remote set-url origin https://github.com/VortexVector/<your-repo-name>.git
git push -u origin main
```
```

---

## Antigravity AI (Pair Programmer)

That screenshot captures the starter code bugs:

- **Anti-diagonal** has 3 X's (``top-right``, ``center``, ``bottom-left``), but the game completely missed the diagonal win.
- Because all 9 squares were filled, it falsely declared **Draw!** Press R for a new round instead of announcing **X wins!**.
- It still displayed **Computer's turn (O)** even though the round had ended.

\*(Save your screen recording or screenshot of this for your **"Before"** video/deliverable).\*

---

### ### Now Run the Fixed ("After") Code

Run this command in your PowerShell:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe\tic-tac-toe"
py -3.13 main.py
```
```

#### What to demonstrate in your 10-second "After" recording:

1. **Diagonal Win**: Complete a diagonal with X `$\rightarrow$` Observe the **X wins!** banner and scoreboard updating to **X: 1**.
2. **Move Validation**: Click on any already-occupied cell `$\rightarrow$` Observe that it is **rejected** and does not overwrite.
3. **Round Restart**: Press **R** `$\rightarrow$` Board clears for a new round, scoreboard **keeps** **X: 1**.
4. **Starter Choice**: Press **T** `$\rightarrow$` Starter switches to **O** and the computer automatically makes the first move.
5. **Match Reset**: Press **M** `$\rightarrow$` Resets the match and resets scoreboard to **O | O | O**.

---

## User

k wt after taking video

---

## Antigravity AI (Pair Programmer)

Once you finish taking the videos, follow these 3 final steps:

---

#### Step 1: Put the 2 Videos into the ``Lab-4/Videos`` folder

Move or copy your two recordings (``before.mp4`` and ``after.mp4``) into:

---

```
c:\Users\rchuk\Downloads\09_tic_tac_toe\Lab-4\Videos\
```

---

\*(Your Windows screen recordings are typically saved in ``Videos\Captures``)\*.

---

#### Step 2: Create a Repo on Your GitHub

1. Open your browser and go to: **[https://github.com/new](https://github.com/new)**
2. Repository name: **09\_tic\_tac\_toe** (or your course lab repository)
3. Set to **Public**
4. Click **Create repository** (do not initialize with README, as we already have code)\*

---

#### Step 3: Push to Your Personal Repo

In your PowerShell terminal, run:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe"
```

# Stage the videos and commit

```
git add .
git commit -m "Add demo videos and complete Lab 4 deliverables"

# Point to your personal GitHub repo and push
git remote set-url origin https://github.com/VortexVector/09_tic_tac_toe.git
git branch -M main
git push -u origin main
```
```

\*(Note: Remember **not** to create a Pull Request to `SETAPESU26`, as stated in Rule 10 of the instructions).\*

---

### Verification: Your `Lab-4` Deliverables  
All deliverables required by the lab rubric are now in  
[`c:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4`](file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/Lab-4):  
- **a. Videos**: `Lab-4/Videos/before.mp4` & `after.mp4`  
- **b. Updated Code**: `Lab-4/Updated\_Code`  
- **c. Chat History**: `Lab-4/PES1UG24CS135\_Lab4\_Chat\_History.pdf` & `.docx`

User

wt nxt

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:15:08+05:30  
Completed At: 2026-09-30T12:15:17+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\Downloads\09\_tic\_tac\_toe\Lab-4\Videos

| Mode   | LastWriteTime    | Length | Name              |
|--------|------------------|--------|-------------------|
| -a---- | 30-09-2026 12:06 | 258    | README_VIDEOS.txt |

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:15:26+05:30  
Completed At: 2026-09-30T12:15:30+05:30

The command exited with code 0.  
Output:

| FullName                                                                 | Length                     | LastWriteTime     |
|--------------------------------------------------------------------------|----------------------------|-------------------|
| C:\Users\rchuk\Videos\test_hardware_encoder.mp4                          | 0                          | 27-08-...         |
| C:\Users\rchuk\Videos\Captures\Orange_ODD_SRN[1].docx - Read-Only - Word | 13-10-2025 10_58_56 AM.png | 242059 13-10-...  |
| C:\Users\rchuk\Videos\Screen Recordings\Screen Recording 2024-12-08      | 194454.mp4                 | 763413 08-12-...  |
| C:\Users\rchuk\Videos\Screen Recordings\Screen Recording 2026-09-30      | 121049.mp4                 | 1939691 30-09-... |

Antigravity AI (Pair Programmer)



Created At: 2026-09-30T12:15:35+05:30  
Completed At: 2026-09-30T12:15:41+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\Videos\Screen Recordings

| Mode   | LastWriteTime    | Length  | Name                                   |
|--------|------------------|---------|----------------------------------------|
| ----   | -----            | -----   | ----                                   |
| -a---- | 08-12-2024 19:44 | 763413  | Screen Recording 2024-12-08 194454.mp4 |
| -a---- | 30-09-2026 12:10 | 1939691 | Screen Recording 2026-09-30 121049.mp4 |

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:15:53+05:30  
Completed At: 2026-09-30T12:15:58+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\Videos\Screen Recordings

| Mode   | LastWriteTime    | Length  | Name                                   |
|--------|------------------|---------|----------------------------------------|
| ----   | -----            | -----   | ----                                   |
| -a---- | 30-09-2026 12:10 | 1939691 | Screen Recording 2026-09-30 121049.mp4 |

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:16:06+05:30  
Completed At: 2026-09-30T12:16:52+05:30

The command exited with code 0.  
Stdout:

Stderr:

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:16:59+05:30  
Completed At: 2026-09-30T12:17:00+05:30

The command exited with code 0.  
Output:  
On branch main  
Your branch is up to date with 'origin/main'.

Untracked files:  
(use "git add <file>..." to include in what will be committed)  
Lab-4/Videos/before.mp4

nothing added to commit but untracked files present (use "git add" to track)

### Antigravity AI (Pair Programmer)

I automatically found your recording in Windows Screen Recordings and copied it to `Lab-4/Videos/before.mp4`.

Now there is only **one last quick step**: record the 10-second **"After"** video.

---

### 1. Run the fixed game

In your PowerShell terminal:

```
```powershell
cd "c:\Users\rchuk\Downloads\09_tic_tac_toe\tic-tac-toe"
py -3.13 main.py
```
```

### 2. Record for 10 seconds (Win + Alt + R or Snipping Tool record):

- Make a diagonal line with X `$\rightarrow` shows **X wins!** and score updates to **X: 1**.
- Click an already filled square `$\rightarrow` shows it doesn't overwrite.
- Press **R** `$\rightarrow` new round starts, keeping score.
- Press **T** `$\rightarrow` toggles starter to O (computer moves first).

### 3. Stop recording and reply here

Once you stop the recording, just reply **"done"**, and I will automatically:

1. Locate and copy the new video to `Lab-4/Videos/after.mp4`.
2. Commit both videos.
3. Push everything directly to your GitHub repository (`VortexVector/09_tic_tac_toe`).

---

User

gimme chat history

---

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:22:28+05:30  
Completed At: 2026-09-30T12:22:28+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\Videos\Screen Recordings

| Mode   | LastWriteTime    | Length  | Name                                   |
|--------|------------------|---------|----------------------------------------|
| ----   | -----            | -----   | -----                                  |
| -a---- | 08-12-2024 19:44 | 763413  | Screen Recording 2024-12-08 194454.mp4 |
| -a---- | 30-09-2026 12:10 | 1939691 | Screen Recording 2026-09-30 121049.mp4 |

---

Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:22:36+05:30  
Completed At: 2026-09-30T12:28:30+05:30

The command exited with code 0.  
Output:

Directory: C:\Users\rchuk\gemini\antigravity\brain\797e1d1a-07d3-4d1b-a401-45de60e4324a\system\_generated\logs

| Mode | LastWriteTime | Length | Name |
|------|---------------|--------|------|
|------|---------------|--------|------|

-----  
d---- 30-09-2026 11:54 chunks  
-a---- 30-09-2026 12:22 147697 transcript.jsonl  
-a---- 30-09-2026 12:22 174265 transcript\_full.jsonl

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:28:40+05:30

Completed At: 2026-09-30T12:28:45+05:30

The command exited with code 1.

Output:

File "<string>", line 8

```
print(f'Type: {stype}, Source: {source}, Content len: {len(obj.get(" content",  
^
```

SyntaxError: unterminated string literal (detected at line 8)

---

### Antigravity AI (Pair Programmer)

Created At: 2026-09-30T12:28:56+05:30

Completed At: 2026-09-30T12:28:56+05:30

Created file file:///c:/Users/rchuk/Downloads/09\_tic\_tac\_toe/export\_chat\_history.py with requested content.

If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.

---